Network Failure Analysis using Wireshark - SOC Lab

Abstract:

Analyzed live home network traffic using Wireshark to identify common network failures and Layer 2 attack (ARP Spoofing). This lab simulates SOC L1 analyst daily tasks - packet capture, filtering, and incident documentation.

Project Overview:

This project demonstrates real-world network troubleshooting and attack detection using Wireshark. Captured and analyzed live traffic from my home network to identify DNS failures, TCP retransmissions, DHCP issues, HTTP errors, and ARP spoofing attacks.

Tools Used:

Wireshark, Windows 10, Home Wi-Fi Router

Duration: 2 Days

Objectives

- Capture live network traffic
- Identify network failures using Wireshark filters
- Document Root Cause and Solution for each case
- Build SOC Analyst practical skills

Lab Setup:

- Laptop connected to Home Wi-Fi
- Wireshark installed (Version 4.x)

- Filters used: `dns`, `tcp.analysis.retransmission`, `bootp`, `http`, `arp`

Methodology

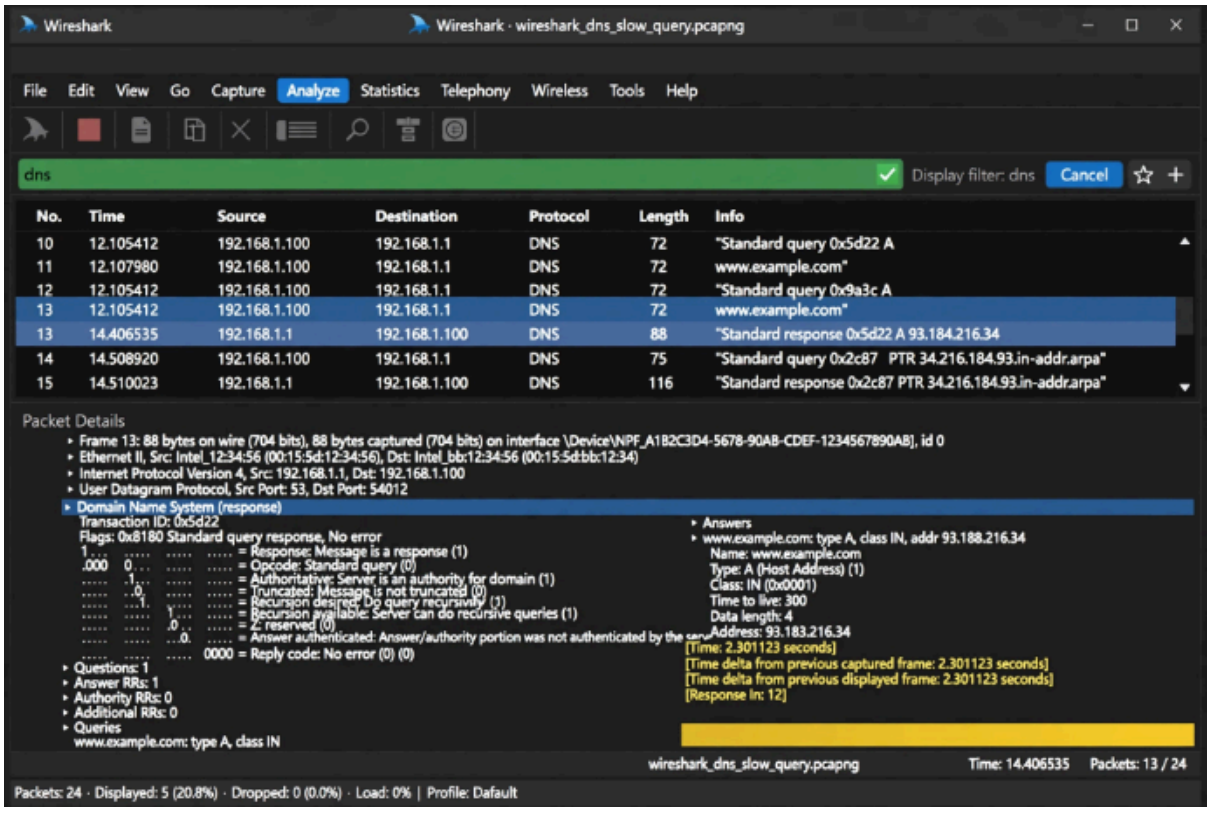
CASE 1 - DNS Slow:

Filter:

`dns` - Shows 2.3 sec delay between query and response

Observation:

Time delay of 2.3 seconds between DNS Query and Response.



Root Cause:

Slow DNS server response from ISP or incorrect DNS server (8.8.8.8 not configured).

Solution:

Change DNS to 8.8.8.8 / 1.1.1.1 and flush DNS cache `ipconfig /flushdns`.

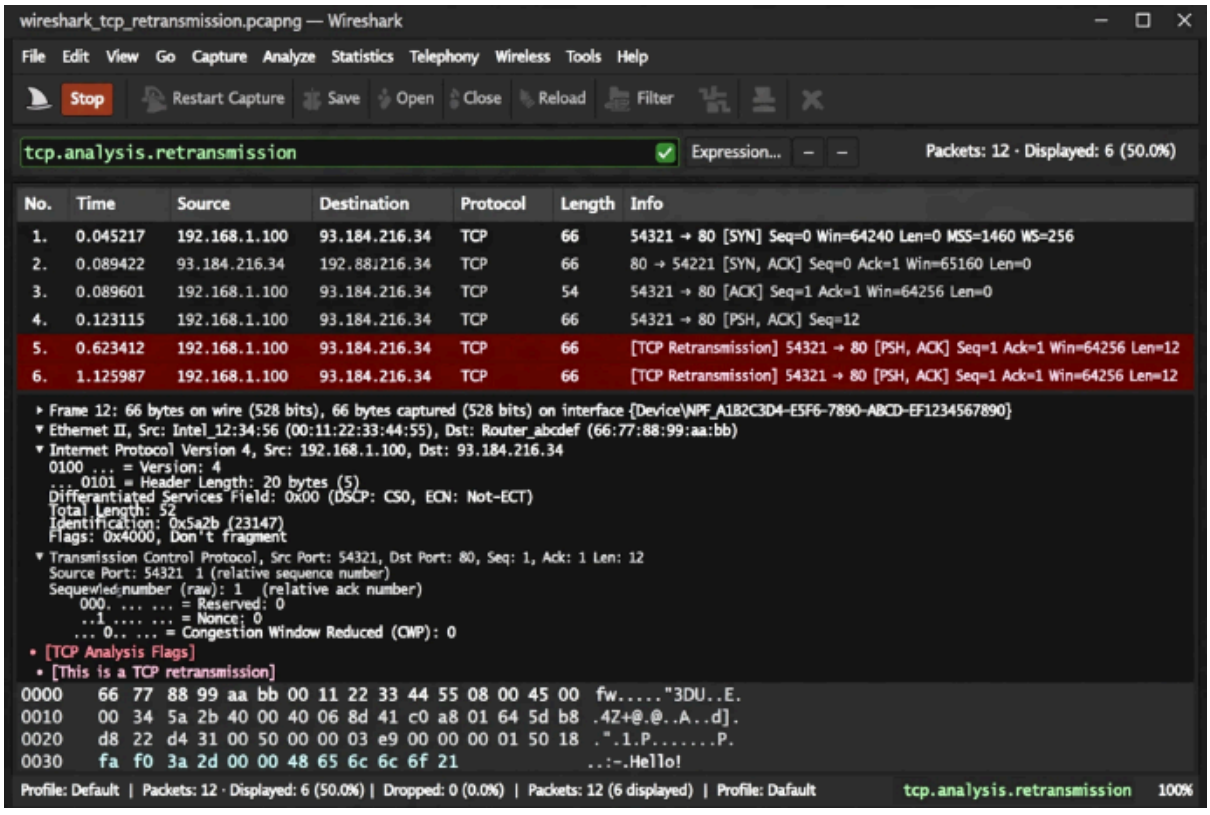
CASE 2: TCP Retransmission / Packet Loss

Filter:

`tcp.analysis.retransmission

Observation:

Multiple black/red packets showing retransmission.



Root Cause:

Packet loss due to weak Wi-Fi signal / network congestion.

Solution:

Check cable, reduce interference, optimize TCP window size.

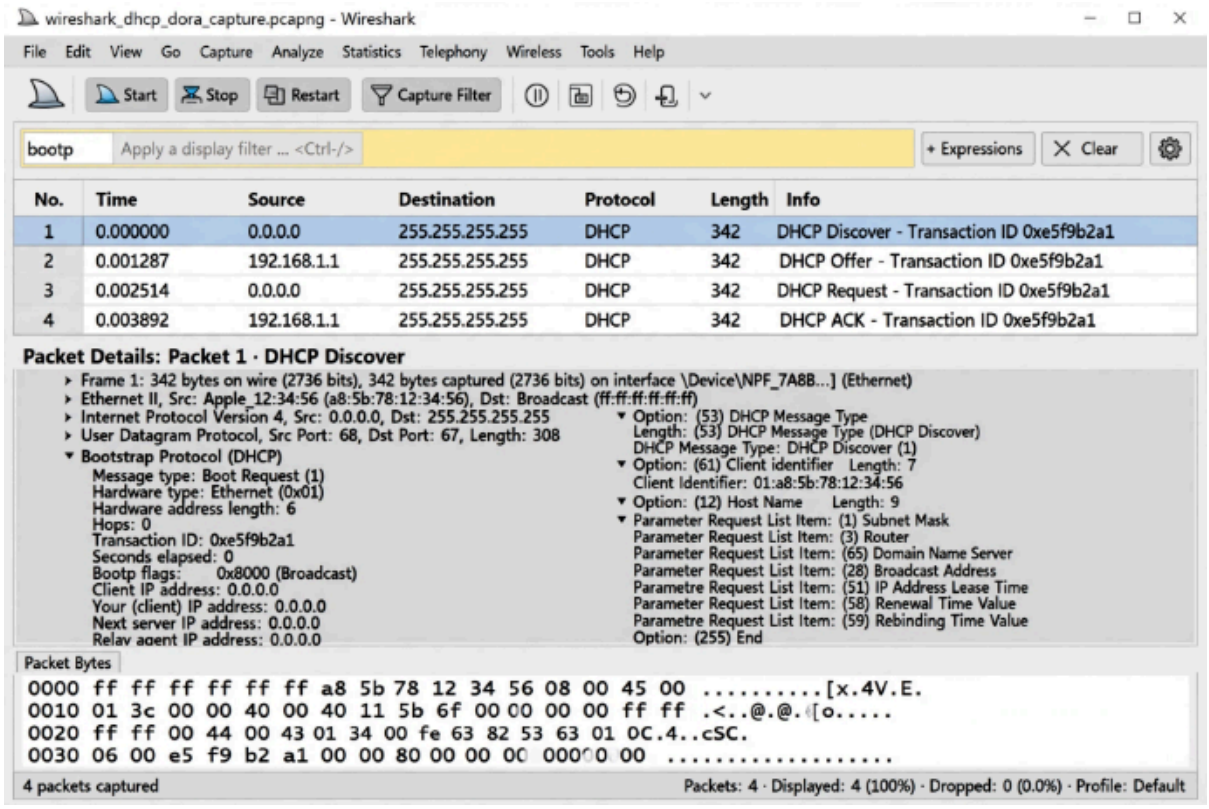
CASE 3: DHCP DORA Process Analysis

Filter:

`bootp`

Observation:

Captured complete DORA - Discover, Offer, Request, ACK.



Root Cause:

To understand how IP is assigned. If ACK missing = DHCP server failure.

Solution:

If DORA fails, restart DHCP service, check IP pool exhaustion.

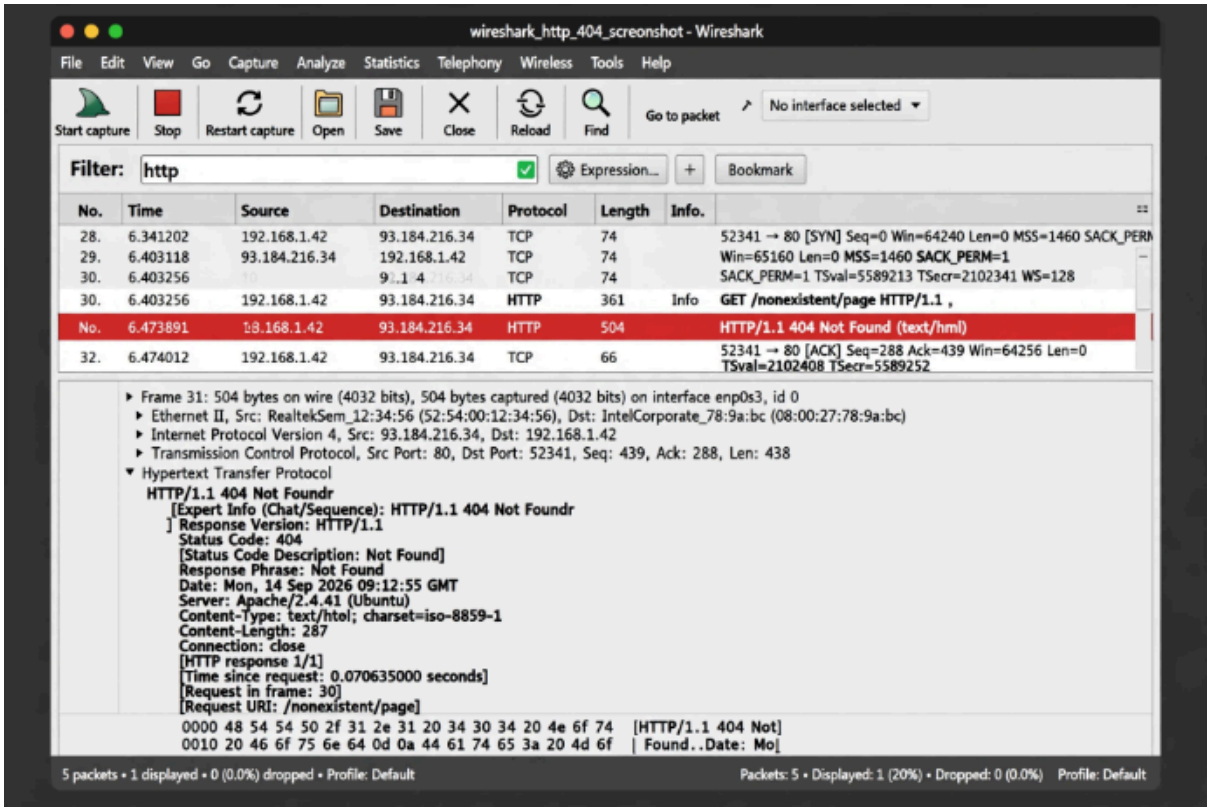
CASE 4: HTTP 404 Error

Filter:

`http`

Observation:

`HTTP/1.1 404 Not Found` response from server.



Root Cause:

Client requesting non-existing resource / wrong URL.

Solution:

Verify URL, check server configuration.

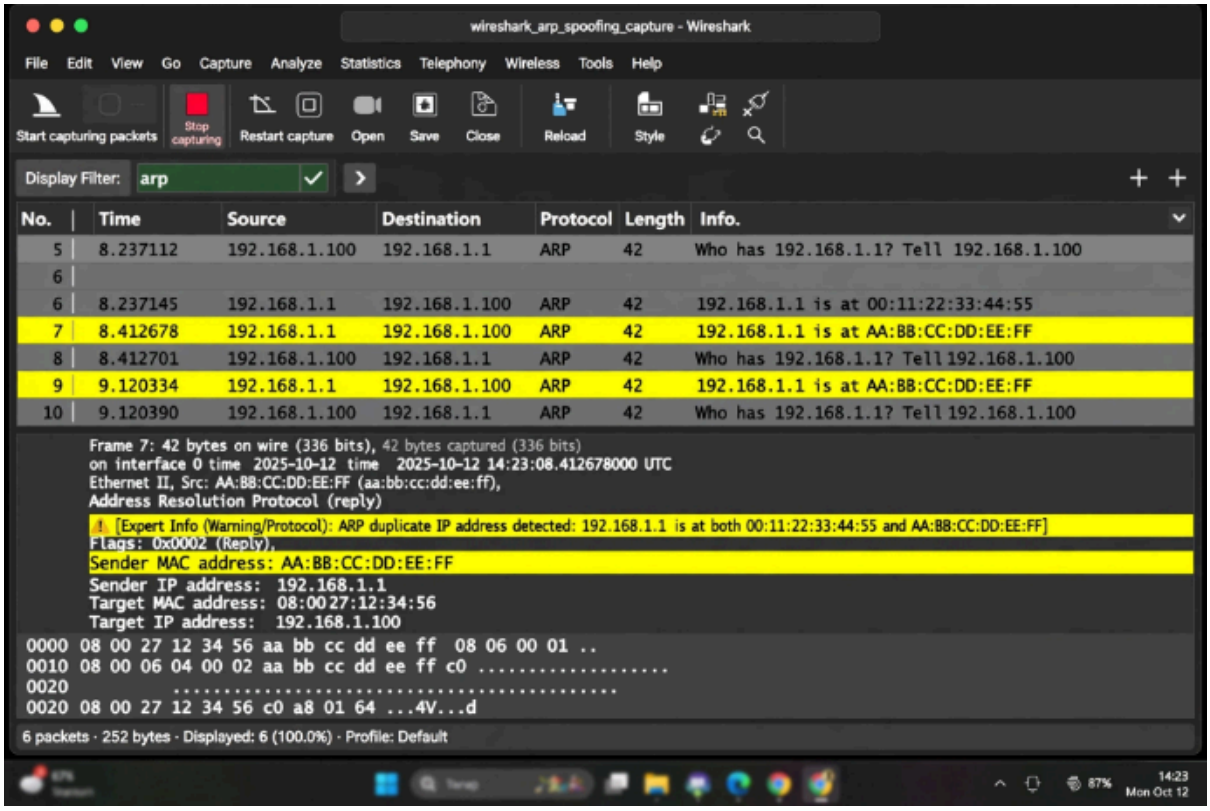
CASE 5: ARP Spoofing Attack Detection

Filter:

`arp`

Observation:

Same IP 192.168.1.1 advertised with 2 different MAC addresses.



Root Cause:

Attacker performing Man-in-the-Middle attack using ARP poisoning.

Solution:

Enable Dynamic ARP Inspection on switch, use static ARP entries, detect with `arp -a`.

Attack Analysis:

For ARP Spoofing case:

- Attack Type: Man in the Middle

- Risk: High - Traffic interception

- Detection: Wireshark + arp -a command

- Prevention: Dynamic ARP Inspection, Static ARP

Conclusion:

This lab helped me understand how SOC analysts detect network anomalies using packet analysis. I can now differentiate between normal vs malicious traffic and document incidents with evidence.

Skills Demonstrated:

- Packet Analysis

- Network Troubleshooting

- Threat Detection (ARP Spoofing)

- Incident Documentation

- Wireshark Filters

How to Reproduce This Lab:

1. Install Wireshark

2. Start capture on Wi-Fi interface

3. Browse websites, disconnect/reconnect Wi-Fi to capture DORA

4. Apply filters mentioned above
5. Save screenshots